

M^5 Day 6: Quadratics

Name: _____

1. Find all real numbers x that satisfy the following equations:

a. $x^2 - 2x - 15 = 0$.

b. $3x^2 + 7x - 4 = 0$

c. $3x^4 + 7x^2 - 4 = 0$

d. $x^{2n} - 2x^n + 1 = 0$ where n is a positive integer.

e. $\sqrt{x+4} = x$

2. Give an example of a quadratic equation $ax^2 + bx + c$ with a, b, c real and $a \neq 0$ such that

a. The equation has no real roots.

b. The equation has exactly one real root, and that root is 5.

c. The equation has roots at $x = 1$ and $x = 0$.

3. This is a “proof” that $2 = 1$. Where is the error?

Let $x = y$. Thus $x^2 = xy$.
Subtracting y^2 from both sides, $x^2 - y^2 = xy - y^2$.
Factoring, $(x + y)(x - y) = y(x - y)$.
Dividing by $(x - y)$ on both sides, $(x + y) = y$.
Therefore $(x + x) = x$.
Divide both sides by x to find that $2 = 1$.